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United States Patent Application Publication

20250286743

Kind Code

A1

Publication Date

September 11, 2025

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PROCESSING SYSTEM AND PROCESSING METHOD

Abstract

A processing system includes: a first storage unit configured to store part of a plurality of processing procedures for processing a packet received via a network; a second storage unit configured to store information on the processing procedures and attribute information related to a power reduction effect in association with each other; and a first processor configured to execute, when a new processing procedure is not able to be added to the first storage unit due to a capacity shortage of the first storage unit, update processing of deleting one of the processing procedures the power reduction effect provided by which is lowest based on the attribute information and then adding the new processing procedure to the first storage unit.

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Appl. No.: 18/779828

Filed: July 22, 2024

Foreign Application Priority Data

JP	2024-034364	Mar. 06, 2024
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Publication Classification

Int. Cl.: H04L12/10 (20060101); H04L12/12 (20060101)

U.S. Cl.:

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based on and claims priority under 35 USC 119 from Japanese Patent Application No. 2024-034364 filed Mar. 6, 2024.

BACKGROUND

(i) Technical Field

[0002] The present disclosure relates to a processing system and a processing method.

(ii) Related Art

[0003] Japanese Unexamined Patent Application Publication No. 2020-205532 discloses the following information processing apparatus. Specifically, the information processing apparatus includes: a power supply control unit that switches between a first power supply mode and a second power supply mode; a main control unit that executes, under the first power supply mode, predetermined processing in accordance with a received packet acquired via a network; and a communication unit that executes processing of transferring the received packet to the main control unit under the first power supply mode and executes processing of discarding the received packet or transferring the received packet to the main control unit under the second power supply mode. The communication unit includes a setting management unit that manages a discard determination pattern which is information for determining whether to discard the received packet or to transfer the received packet to the main control unit. The main control unit includes a setting change unit that changes a list of the discard determination patterns managed by the setting management unit, based on determination on whether processing on the received packet in the main controller is discard processing.

SUMMARY

[0004] Aspects of non-limiting embodiments of the present disclosure relate to a processing system and a processing method enabling a processing procedure with a high power reduction effect to be retained when a new processing procedure is added to a first storage unit.

[0005] Aspects of certain non-limiting embodiments of the present disclosure address the above advantages and/or other advantages not described above. However, aspects of the non-limiting embodiments are not required to address the advantages described above, and aspects of the non-limiting embodiments of the present disclosure may not address advantages described above.

[0006] According to an aspect of the present disclosure, there is provided a processing system including: a first storage unit configured to store part of a plurality of processing procedures for processing a packet received via a network; a second storage unit configured to store information on the processing procedures and attribute information related to a power reduction effect in association with each other; and a first processor configured to execute, when a new processing procedure is not able to be added to the first storage unit due to a capacity shortage of the first storage unit, update processing of deleting one of the processing procedures the power reduction effect provided by which is lowest based on the attribute information and then adding the new processing procedure to the first storage unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] Exemplary embodiments of the present disclosure will be described in detail based on the

following figures, wherein:

[0008] FIG. 1 is a block diagram illustrating a hardware configuration of an information processing system according to a first exemplary embodiment;

[0009] FIG. 2 is a block diagram illustrating an example of a functional configuration of a main processing unit in the information processing system;

[0010] FIG. 3 is a block diagram illustrating an example of a functional configuration of a communication unit in the information processing system;

[0011] FIG. 4 is a diagram illustrating an example of a proxy response pattern;

[0012] FIG. 5(A) is a diagram illustrating an example of a setting screen for setting attribute information displayed on an operation panel, and FIG. 5(B) is a diagram illustrating an example of a setting report in which the attribute information is set;

[0013] FIG. 6 is a flowchart illustrating a flow of update determination processing executed by the main processing unit of the information processing system according to the first exemplary embodiment;

[0014] FIG. 7 is a flowchart illustrating a flow of processing for measuring a packet processing time executed by the main processing unit of the information processing system according to the first exemplary embodiment; and

[0015] FIG. 8 is a flowchart illustrating a flow of processing executed by the communication unit of the information processing system according to the first exemplary embodiment to accumulate usage records.

DETAILED DESCRIPTION

[0016] Hereinafter, an example of an exemplary embodiment according to the present disclosure will be described with reference to the drawings. Note that in the drawings, the same or equivalent components and parts are denoted by the same reference signs. In addition, dimensional ratios in the drawings are exaggerated for convenience of description and may be different from actual ratios.

First Exemplary Embodiment

[0017] FIG. 1 is a block diagram illustrating a hardware configuration of an information processing system 2 as an example of a processing system according to a first exemplary embodiment.

Hardware Configuration of Information Processing System

[0018] For example, the information processing system 2 includes a main processing unit 10, an operation panel 20, an image forming unit 22, a scan unit 24, a fax machine 26, and a communication unit 30. Thus, the information processing system 2 includes the image forming unit 22, the scan unit 24, the fax machine 26, and the communication unit 30, to be configured as a multifunction peripheral having functions of those of a printer, a copier, and a facsimile, as well as e-mail transmission and the like. Alternatively, the information processing system 2 may be applied to a computer apparatus such as a desktop computer, a laptop computer, or a tablet computer having a mail transmission function. The “system” in the present disclosure includes both a system configured by a plurality of apparatuses and a system configured by a single apparatus.

[0019] As illustrated in FIG. 1, the main processing unit 10 includes the following components: a central processing unit (CPU) 11, a read only memory (ROM) 12, a random access memory (RAM) 13, a storage 14, and an input/output interface 15. The components are communicably connected to each other via a bus 19.

[0020] The CPU 11 is a central processing unit, and executes various programs and controls each unit. The CPU 11 is an example of a first processor. The CPU 11 reads a program from the ROM 12 or the storage 14, and executes the program using the RAM 13 as a work area. In the present exemplary embodiment, a processing program is stored in the ROM 12 or the storage 14. The CPU 11 controls the above-described components and performs various kinds of arithmetic processing in accordance with the processing program stored in the ROM 12 or the storage 14.

[0021] The ROM 12 stores therein various programs and various types of data. The RAM 13,

serving as a work area, temporarily stores therein programs or data. The storage **14** includes a hard disk drive (HDD) or a solid state drive (SSD), and stores various programs including an operating system and various types of data.

[0022] The input/output interface **15** is an interface for transmitting and receiving information to and from peripheral devices of the main processing unit **10**. The input/output interface **15** is connected to each of the operation panel **20**, the image forming unit **22**, the scan unit **24**, the fax machine **26**, and the communication unit **30**, and can transmit and receive information to and from each of these components. Note that in the information processing system **2** according to the present disclosure, instead of this configuration, the input/output interface **15** may be directly connected to at least one of the operation panel **20**, the image forming unit **22**, the scan unit **24**, the fax machine **26**, and the communication unit **30** via the bus **19**.

[0023] The operation panel **20** has a function of a display unit which displays various types of information and a function of an input unit which is operated by a user. As an example, the operation panel **20** is a touch panel user interface (UI), and is a touch panel display having a touch panel superimposed on a liquid crystal display.

[0024] The image forming unit **22** acquires image data and forms an image on a recording medium. The image forming unit **22** executes, for example, processing for conveying a recording medium to an image forming position, processing for forming an image on the recording medium, processing for ejecting the discharging the recording medium on which the image has been formed, and the like. Specifically, the image forming unit **22** includes an electrophotographic image forming mechanism, a recording medium conveying mechanism, and the like, and upon receiving image data indicating an image to be printed and various operation commands from the main processing unit **10**, performs printing on one side or both sides of the recording medium and then discharges the recording medium to an external paper output tray.

[0025] The scan unit **24** conveys a plurality of documents and reads each of the documents. Specifically, the scan unit **24** executes processing for conveying a document to a reading position, processing for reading the document, processing for discharging the document, and the like. When an operation command for reading a document is input from the main processing unit **10**, the scan unit **24** reads the document, acquires image data which is the read image, and discharges the document to a discharge portion.

[0026] The fax machine **26** performs management for a destination to which image data or the like is transmitted, transmission and reception of the image data, and the like. For example, the fax machine **26** executes processing for transmitting image data read by the scan unit **24** to a designated destination or receiving image data transmitted from the outside. The fax machine **26** has, for example, a communication function for transmitting and receiving image data and the like. Note that alternatively, the fax machine **26** may transmit and receive image data and the like via the communication unit **30**.

[0027] The communication unit **30** has an information communication function. The communication unit **30** includes components including a central processing unit (CPU) **31**, a read only memory (ROM) **32**, a random access memory (RAM) **33**, a storage **34**, and a communication interface **35**. The components are communicably connected to each other via a bus **39**. The communication unit **30** performs packet communications, for example, over a network **40** described later. A packet refers to a unit of transmission in a case where data is divided into small pieces data of a specific size to be sent in information communication, and refers to a mass of data exchanged over a network. For example, when communication is performed for large amount of data, the data is divided. Then, each of the divided pieces of data is provided with control information such as an address and is communicated.

[0028] The CPU **31** is a central processing unit of the communication unit **30**, and executes various programs and controls the components in the communication unit **30**. The CPU **31** is an example of a second processor.

[0029] The ROM **32** stores therein various programs and various types of data. The RAM **33** serves as a work area and temporarily stores therein programs or data. The storage **34** is configured by an HDD or an SSD and stores various programs and various types of data.

[0030] The communication interface **35** is an interface for communicating with the network **40**, using for example, a standard such as Ethernet (registered trademark), FDDI, or Wi-Fi (registered trademark). The network **40** is, for example, the Internet, or a wired or wireless network. As an example, the information processing system **2** is connected to a server (not illustrated) via the network **40**, and can perform information communication with the server.

Functional Configuration of Main Processing Unit

[0031] FIG. **2** is a block diagram illustrating an example of a functional configuration of the main processing unit **10** in the information processing system **2**.

[0032] As illustrated in FIG. **2**, the main processing unit **10** includes, as functional configurations, an attribute information changing unit **51**, a storage unit **52**, an update determination unit **53**, an information output unit **54**, an information acquisition unit **55**, a packet processing unit **56**, and a measurement unit **57**. Each of the functional configurations is realized by the CPU **11** reading an information processing program stored in the ROM **12** or the storage **14** and loading and executing the information processing program on the RAM **13**.

[0033] The attribute information changing unit **51** changes attribute information **62**, which is information on the attribute of a pattern. For example, the attribute information changing unit **51** changes the attribute information **62** based on the information acquired from the communication unit **30**.

[0034] The storage unit **52** stores the attribute information **62** and pattern information **64** which is information on a plurality of patterns. The storage unit **52** is an example of a second storage unit. Furthermore, the plurality of patterns are an example of a plurality of processing procedures for processing packets received via the network **40**. That is, a pattern is an example of a processing procedure. The pattern information **64** is information for identifying a plurality of patterns and is not the patterns themselves. The pattern information **64** includes, for example, ID information of the pattern and the like. The ID information is a code distinctively assigned by an information device or service provider to each of the patterns.

[0035] In the first exemplary embodiment, the RAM **13** of the main processing unit **10** or the storage **14** functions as the storage unit **52**. The storage unit **52** preferably has a relatively long service life and a relatively large storage capacity. The storage unit **52** includes, for example, a nonvolatile semiconductor memory (flash memory) stored contents in which are retained even when the power is turned OFF.

[0036] As described above, the attribute information **62** is information on the attribute of the pattern. The attribute information **62** includes information on a power reduction effect of a pattern. The attribute information **62** includes, for example, first attribute information **62A**, second attribute information **62B**, third attribute information **62C**, and fourth attribute information **62D**. The attribute information **62** is not limited to the first attribute information **62A**, the second attribute information **62B**, the third attribute information **62C**, and the fourth attribute information **62D**, and may include other types of attribute information.

[0037] The first attribute information **62A** is, for example, rewritability information of a pattern indicating whether the pattern is rewritable. The pattern rewritability information can be set by, for example, a user or a developer of the pattern.

[0038] The second attribute information **62B** is, for example, the usage record count of the pattern. For example, when the pattern is used on the communication unit **30** side in an energy-saving state as will be described later, the information on the usage record count is temporarily stored in the RAM **33** on the communication unit **30** side, and the information on the usage record count is read from the communication unit **30** at the time of return from the energy-saving state to be stored in the storage unit **52**.

[0039] The third attribute information **62C** is, for example, the packet processing time. For example, when the packet processing unit **56** processes a packet, the measurement unit **57** measures the processing time of the packet and stores the measured processing time of the packet as the third attribute information **62C**. Note that the packet processing time may have an initial value set in advance.

[0040] The fourth attribute information **62D** is, for example, expected power reduction effect information. The expected power reduction effect information is calculated as the product of the usage record count of the pattern, which is the second attribute information **62B**, and the packet processing time, which is the third attribute information **62C**.

[0041] FIG. **4** illustrates an example of a table in which a proxy response pattern and the attribute information **62** are associated with each other. For example, the table illustrated in FIG. **4** is stored in the storage unit **52**. As illustrated in FIG. **4**, the pattern of the third example is not unrewritable (that is, rewritable), and the power reduction effect, which is a numerical value of the fourth attribute information **62D**, provided therewith is the lowest.

[0042] The update determination unit **53** determines whether to cause the communication unit **30** to perform update processing for a new pattern. For example, the update determination unit **53** determines that the new pattern can be stored in the storage unit **72** when a storage unit **72** (see FIG. **3**) of the communication unit **30** has a sufficient capacity, or that the new pattern cannot be stored in the storage unit **72** when the storage unit **72** (see FIG. **3**) of the communication unit **30** does not have a sufficient capacity. When the new pattern cannot be added to the storage unit **72** due to a capacity shortage of the storage unit **72** of the communication unit **30**, for example, the main processing unit **10** (for example, the CPU **11**) causes the communication unit **30** to execute update processing of deleting a pattern with the lowest power reduction effect based on the attribute information **62** and then adding the new pattern to the storage unit **72**.

[0043] The update determination unit **53** selects the deletion target pattern with the lowest power reduction effect based on the attribute information **62** (for example, the usage record count of the pattern which is the second attribute information **62B** and the pattern processing time which is the third attribute information **62C**). For example, as illustrated in FIG. **4**, the pattern of the third example is not unrewritable (that is, rewritable), and the power reduction effect, and the power reduction effect, which is a numerical value of the fourth attribute information **62D**, provided therewith is the lowest. Therefore, the pattern of the third example is selected as the deletion target at the time of the next update.

[0044] When the pattern rewritability information which is the first attribute information **62A** indicates that the pattern is unrewritable, the update determination unit **53** does not cause the communication unit **30** to delete the pattern in the storage unit **72** regardless of the power reduction effect. For example, as illustrated in FIG. **4**, the first attribute information **62A** indicates that a pattern of a first example is unrewritable. In this case, the pattern of the first example is not deleted.

[0045] For example, the update determination unit **53** acquires a pattern rewrite count, and does not cause the communication unit **30** to execute the update processing of adding the new pattern to the storage unit **72** when the pattern rewrite count is equal to or more than the specified value. Here, the pattern rewrite count indicates the number of times update processing such as deletion of a pattern or addition of a new pattern has been executed. For example, when a pattern is deleted and a new pattern is added, the rewrite count is two. The pattern rewrite count is acquired from the communication unit **30**.

[0046] The information output unit **54** outputs information to the communication unit **30**. For example, the information output unit **54** outputs, to the communication unit **30**, a request to perform the update processing for a new pattern, a request to delete a deletion target pattern, and the like.

[0047] The information acquisition unit **55** acquires information from the communication unit **30**. For example, information such as the pattern rewrite count is acquired from the communication

unit **30**.

[0048] The packet processing unit **56** processes a packet. In one example, when the packet is not processed by a discard pattern **82A** or a proxy response pattern **82B** (refer to FIG. **3**) in the communication unit **30**, the packet is processed by the packet processing unit **56** in the main processing unit **10**.

[0049] The measurement unit **57** measures a packet processing time. The measurement unit **57** includes a timer and measures the packet processing time using the timer. For example, when a packet is the processing time measurement target, the measurement unit **57** measures the processing time for the packet. The packet processing time is stored in the storage unit **52** as the third attribute information **62C**. The packet as the processing time measurement target is a packet involving no job. Examples of the job include printing on a recording medium by the image forming unit **22** and reading of a document by the scan unit **24**. In other words, the packet involving no job is a packet for which only packet processing is executed and no job is performed. Since the packet involving no job is a target of filtering or an automatic response in the communication unit **30**, the processing time of such a packet is accumulated in the third attribute information **62C**.

Setting Screen and Setting Report for Attribute Information

[0050] FIG. **5(A)** illustrates an example of a setting screen for the user to set the attribute information, and FIG. **5(B)** illustrates an example of a setting report for the user to check the attribute information **62**. As illustrated in FIG. **5(A)**, a setting screen **21A** for setting the attribute information **62** is displayed on the operation panel **20**. On the setting screen **21A**, “permit dynamic change of return factor” is set to “no” (thus, rewriting is prohibited). A black frame portion **100** in the setting screen **21A** is a new portion. Thus, as the first attribute information **62A** set by the user, the pattern can be set to be unrewritable. Note that unlike in the above-described setting screen **21A**, the unrewritable part may be grayed out.

[0051] Further, as shown in FIG. **5(B)**, a setting report **21B** for checking the setting of the attribute information is output to the recording medium by the image forming unit **22**. Alternatively, the setting report **21B** may be displayed on the operation panel **20**. A black frame portion **102** in the setting report **21B** is a new portion. In the setting report **21B**, “permit dynamic change of return factor” is set to “no” (thus, rewriting is prohibited). Thus, it is confirmed that for the pattern, unrewritable is set as the first attribute information **62A**. Further, in the setting report **21B**, as the second attribute information **62B**, it is confirmed that the usage record count of the pattern is 43 times.

Functional Configuration of Communication Unit

[0052] FIG. **3** is a block diagram illustrating an example of a functional configuration of the communication unit **30** in the information processing system **2**.

[0053] As illustrated in FIG. **3**, the communication unit **30** includes, as functional configurations, a pattern update processing unit **71**, a storage unit **72**, a packet processing unit **73**, an information output unit **74**, an information acquisition unit **75**, a transmission unit **76**, and a reception unit **77**. Each of the functional configurations is realized by the CPU **31** reading an information processing program stored in the ROM **32** or the storage **34** and loading and executing the information processing program on the RAM **33**.

[0054] The pattern update processing unit **71** executes pattern update processing. The pattern update processing includes processing of deleting a pattern, adding a new pattern, and the like. For example, the pattern update processing unit **71** executes the pattern update processing upon acquiring a request to execute the pattern update processing from the main processing unit **10**. The pattern on which the update processing has been executed is stored in the storage unit **72**.

[0055] The storage unit **72** stores a plurality of patterns **82**. The storage unit **72** is an example of a first storage unit. The plurality of patterns **82** are an example of a plurality of processing procedures for processing packets received via the network **40**. Examples of the pattern **82** include

the discard pattern **82A**, the proxy response pattern **82B**, and the like. The discard pattern **82A** is, for example, a processing procedure for discarding a packet at the communication unit **30** without activating the main processing unit **10**. The proxy response pattern **82B** is, for example, a processing procedure for causing the communication unit **30** to perform a proxy response for a packet without activating the main processing unit **10**.

[0056] In the first exemplary embodiment, the RAM **33** of the communication unit **30** or the storage **34** functions as the storage unit **72**. The storage unit **72** includes, for example, a nonvolatile semiconductor memory (flash memory) stored contents in which are maintained even when the power is turned OFF. The rewrite count information for the pattern **82** may be stored in the storage unit **72** or may be stored in a memory other than the storage unit **72**.

[0057] The service life of the storage unit **72** regarding the rewrite count is, for example, 100,000 times in the case of a flash ROM. A flash ROM (flash memory) is a type of electronically erasable and programmable read only memory (EEPROM), and is a ROM in which contents can be electrically erased and written in block units. In the storage unit **72**, a specified value of the rewrite count of the pattern **82** is set in advance, and when the rewrite count of the pattern **82** in the storage unit **72** is equal to or more than the specified value, the main processing unit **10** (for example, the CPU **11**) performs control to prohibit the rewrite of the pattern **82**. The specified value is a value obtained by subtracting a margin from the service life of the storage unit **72** regarding the rewrite count (for example, 100,000 times in the case of a flash ROM), and for example, when the margin is 20%, the specified value is 80,000 times, which is 80% for example. As the storage unit **72**, it is preferable to use a magnetoresistive random access memory (MRAM) or a ferroelectric random access memory (FRAM) (registered trademark) featuring a longer service life than a flash ROM. The MRAM uses a magnetic material as a part of the material of a storage element, and records a signal through a change in a magnetized state. The FRAM is a nonvolatile memory device using a dielectric substance as a storage element.

[0058] The packet processing unit **73** processes a packet. As an example, in a case where the packet received by the communication unit **30** corresponds to the discard pattern **82A** in which the discarding is performed or corresponds to the proxy response pattern **82B** in which the proxy response is performed, the packet processing unit **73** processes the packet without activating (booting) the main processing unit **10**.

[0059] After processing the packet, the packet processing unit **73** stores, in the storage unit **72**, incremental information for incrementing the pattern usage record count in the attribute information **62**, without activating the main processing unit **10**. For example, the incremental information may be temporarily stored in the storage unit **72**, and thereafter, when the main processing unit **10** is activated, the main processing unit **10** may be notified of the incremental information. Upon being notified of the incremental information, the main processing unit **10** stores the pattern usage record count as the second attribute information **62B** in the storage unit **52**.

[0060] The information output unit **74** outputs information to the main processing unit **10**. For example, the information output unit **74** outputs, to the main processing unit **10**, the incremental information for incrementing the pattern usage record count in the attribute information **62**. For example, in a case where the packet does not correspond to the discard pattern **82A**, in which the packet is discarded, or does not correspond to the proxy response pattern **82B**, in which a proxy response is performed for the packet, the pattern usage record count may be output to the main processing unit **10** in response to a request from the main processing unit **10**.

[0061] The information acquisition unit **75** acquires information from the main processing unit **10**. For example, a request for deleting a pattern or executing the update processing for a new pattern is acquired from the main processing unit **10**. For example, when the information acquisition unit **75** acquires the request for deleting the pattern from the main processing unit **10**, the pattern update processing unit **71** executes pattern deletion processing. For example, when the information acquisition unit **75** acquires the request for executing the update processing for a new pattern from

the main processing unit **10**, the pattern update processing unit **71** executes the update processing for a new pattern.

[0062] The transmission unit **76** transmits information to the outside via the network **40**. For example, the transmission unit **76** transmits a packet via the network **40**.

[0063] The reception unit **77** receives information from the outside via the network **40**. For example, the reception unit **77** receives a packet via the network **40**.

Operation According to First Exemplary Embodiment

[0064] Next, the operation of the information processing system **2** according to the first exemplary embodiment will be described.

[0065] FIG. **6** is a flowchart illustrating a flow of the update determination processing executed by the main processing unit **10** of the information processing system **2** according to the first exemplary embodiment. In the main processing unit **10**, the CPU **11** reads an information processing program for the update determination from the ROM **12** or the storage **14**, loads the information processing program on the RAM **13**, and executes the information processing program. Thus, the information processing is executed.

[0066] The CPU **11** determines whether an update event for the pattern **82** has occurred (step **S201**). The update event occurs when the pattern **82** needs to be updated. For example, the pattern **82** needs to be updated when the user changes the setting information of the attribute information **62** using the operation panel **20** or when the attribute information **62** is updated and the rank changes. For example, a case of occurrence of an update event for adding a new pattern to the storage unit **72** of the communication unit **30** will be described.

[0067] When the update event for the pattern **82** has not occurred (step **S201**: NO), the CPU **11** waits until the update event for the pattern **82** occurs.

[0068] When the update event for the pattern **82** occurs (step **S201**: YES), the CPU **11** determines whether the rewrite count for the pattern **82** is equal to or more than the specified value (step **S202**). The rewrite count for the pattern **82** is acquired from the communication unit **30**. As described above, the specified value is a value obtained by subtracting a margin (for example, 20%) from the rewrite count related service life of the storage unit **72** of the communication unit **30**. For example, the CPU **11** calculates the specified value from the rewrite count related service life of the storage unit **72** acquired from the communication unit **30**, and determines whether the rewrite count for the pattern **82** is equal to or more than the specified value.

[0069] When the rewrite count for the pattern **82** is equal to or more than the specified value (step **S202**: YES), the CPU **11** ends the processing without updating the pattern **82** (step **S203**). In other words, since the rewrite count for the pattern **82** has reached the specified value, the pattern **82** is not updated. Thus, the processing executed, based on the information processing program, by the main processing unit **10** of the information processing system **2** ends.

[0070] When the rewrite count for the pattern **82** is less than the specified value (step **S202**: NO), the CPU **11** determines whether the storage unit **72** has a sufficient capacity for adding the pattern **82** (step **S204**). The capacity of the storage unit **72** is acquired from the communication unit **30**.

[0071] When the storage unit **72** has a sufficient capacity for adding the pattern **82** (step **S204**: YES), the CPU **11** outputs an update request for the pattern **82** to the communication unit **30** (step **S205**). Upon receiving the update request for the pattern **82** from the main processing unit **10**, the CPU **31** of the communication unit **30** executes the update processing for the pattern **82** (for example, update processing for adding a new pattern to the storage unit **72**). Thus, the processing executed, based on the information processing program, by the main processing unit **10** of the information processing system **2** ends.

[0072] When the storage unit **72** does not have a sufficient capacity for adding the pattern **82** (step **S204**: NO), the CPU **11** selects the pattern **82** with the lowest power reduction effect other than the rewrite prohibition as the deletion target (step **S206**). For example, as illustrated in FIG. **4**, based on the attribute information **62** (the first attribute information **62A** to the fourth attribute information

62D), the CPU **11** selects, as the deletion target, the pattern **82** (for example, the pattern of the third example illustrated in FIG. **4**) with the lowest power reduction effect other than the rewrite prohibition.

[0073] The CPU **11** outputs an update request together with deletion information to the communication unit **30** (step **S207**). Upon receiving the update request for the pattern **82** together with the deletion information from the main processing unit **10**, the CPU **31** of the communication unit **30** deletes the deletion target pattern (for example, the pattern of the third example illustrated in FIG. **4**) and then executes the update processing for the pattern **82** (for example, the update processing of adding a new pattern to the storage unit **72**). Thus, the processing executed, based on the information processing program, by the main processing unit **10** of the information processing system **2** ends.

[0074] FIG. **7** is a flowchart illustrating a flow of processing of measuring a packet processing time executed by the main processing unit **10** of the information processing system **2** according to the first exemplary embodiment. In the main processing unit **10**, the CPU **11** reads an information processing program for the measurement from the ROM **12** or the storage **14**, loads the program on the RAM **13**, and executes the program. Thus, the information processing is executed.

[0075] The CPU **11** determines whether a return by packet is detected (step **S251**). For example, this corresponds to a case where the packet does not correspond to the discard pattern **82A** or the proxy response pattern **82B** in the communication unit **30**, and is processed by the main processing unit **10** activated (i.e., returned). When a signal for making the main processing unit **10** return is transmitted from the communication unit **30**, the CPU **11** determines that the return by packet is detected.

[0076] When the return by packet is not detected (step **S251**: NO), the CPU **11** waits until the return by packet is detected.

[0077] When the return by packet is detected (step **S251**: YES), the CPU **11** determines whether the packet is a processing time measurement target packet (step **S252**). A packet not requiring the measurement is excluded through the determination on whether the packet is the processing time measurement target packet. For example, the processing time measurement target packet is a packet not involving a job (that is, a packet for which only the packet processing is executed). Since a packet not involving a job is a target of discarding (that is, removal by a filter) or a proxy response by the communication unit **30**, the processing time is measured for such a packet. The non-target packet is a packet involving a job such as printing or scanning

[0078] When the packet is a target of the processing time measurement (step **S252**: YES), the CPU **11** starts measuring the packet processing time and executes the packet processing (step **S253**).

[0079] When the packet is not the target of the processing time measurement (step **S252**: NO), the CPU **11** executes the packet processing without measuring the packet processing time (step **S254**).

[0080] The CPU **11** ends the packet processing time measurement when the packet processing ends, and stores the packet processing time as the third attribute information **62C** (step **S255**).

[0081] After the processing in step **S254** or step **S256**, the CPU **11** ends the processing (step **S256**). Thus, the processing executed, based on the information processing program, by the main processing unit **10** of the information processing system **2** ends.

[0082] FIG. **8** is a flowchart illustrating a flow of processing of accumulating usage records executed by the communication unit **30** of the information processing system **2** according to the first exemplary embodiment. In the communication unit **30**, the CPU **31** reads an information processing program for accumulating usage records from the ROM **32** or the storage **34**, loads the program on the RAM **33**, and executes the program. Thus, the information processing is executed.

[0083] The CPU **31** transitions to an energy-saving state (step **S301**). For example, when the main processing unit **10** transitions to the energy-saving state with the information processing system **2** not operating for a predetermined period of time, the communication unit **30** also transitions to the energy-saving state.

[0084] The CPU **31** receives a packet (step **S302**). In this example, the CPU **31** receives a packet from the outside via the network **40**. The CPU **31** may make the communication unit **30** return from the energy-saving state to the normal state upon receiving a packet.

[0085] The CPU **31** determines whether the packet corresponds to any one of the discard pattern or the proxy response pattern (step **S303**). A packet corresponding to any one of the discard pattern and the proxy response pattern is a packet corresponding to one of the discard pattern **82A** and the proxy response pattern **82B**, and leads to “YES” in step **S303** in this case. A packet corresponding to neither one of the discard pattern **82A** and the proxy response pattern **82B** is a packet not corresponding to the discard pattern **82A** or the proxy response pattern **82B**, and leads to “NO” in step **S303**.

[0086] When the packet corresponds to any one of the discard pattern or the proxy response pattern (step **S303**: YES), the CPU **31** executes specified processing (step **S304**). For example, when the packet corresponds to the discard pattern **82A**, the CPU **31** executes processing of discarding the packet as the specified processing. For example, when the packet corresponds to the proxy response pattern **82B**, the CPU **31** executes processing of performing a proxy response for the packet as the specified processing.

[0087] When the packet corresponds to none of the discard pattern and the proxy response pattern (step **S303**: NO), the CPU **31** boots the main processing unit **10** (step **S305**). For example, the CPU **31** boots the main processing unit **10** by outputting a return by packet signal to the main processing unit **10**.

[0088] After the processing in step **S304**, the CPU **31** increments the information on the usage record count of the pattern **82** (the discard pattern **82A** or the proxy response pattern **82B**) (step **S306**). This incrementation is an operation of adding 1 to the numerical value. While the usage record count of the pattern **82** (the discard pattern **82A** or the proxy response pattern **82B**) is the second attribute information **62B**, the processing in step **S306** is executed without booting the main processing unit **10**. For example, the CPU **31** temporarily stores the incremental information for the usage record count of the pattern **82** in the storage unit **72**.

[0089] For example, the incremental information for the usage record count of the pattern **82** is transmitted to the main processing unit **10** when the main processing unit **10** is booted (returns). Thus, the CPU **11** of the main processing unit **10** stores the usage record count of the pattern **82** in the storage unit **52** as the second attribute information **62B** based on the incremental information received from the communication unit **30**.

[0090] The CPU **31** returns to the processing in step **S301**. Thus, the CPU **31** transitions to the energy-saving state again.

[0091] After the processing in step **S305**, the CPU **31** transmits the usage record count of the pattern **82** based on a request from the main processing unit **10** (step **S307**). The usage record count of the pattern **82** is stored as the second attribute information **62B** in the storage unit **52** of the main processing unit **10**.

[0092] The CPU **31** ends the processing (step **S308**). Thus, the processing based on the information processing program executed by the communication unit **30** of the information processing system **2** ends.

[0093] In general, there is a strong demand for reducing power consumption of a multifunction peripheral having functions such as those of printer, copier, fax machine, as well as e-mail transmission. In order to satisfy this demand, a filtering technique has been proposed for a multifunction peripheral of Comparative Example. According to such a technique, a discard pattern that is a condition for discarding a received packet is set in advance, and when a packet received during power saving control corresponds to the pattern, the packet is discarded by a communication unit without being processed by a main controller (such as a CPU). In addition, as a technique similar to the filter technique, there is a technique in which a proxy response pattern that is a condition for performing proxy response for a received packet is set in advance, and the proxy

response is performed for a predetermined packet corresponding to the pattern.

[0094] However, in the multifunction peripheral of Comparative Example, since the type of the packet received frequently differs depending on the network environment, the power consumption may not be sufficiently reduced by simply setting the predetermined discard pattern or proxy response pattern.

[0095] The information processing system **2** according to the first exemplary embodiment includes the storage unit **72** that stores the plurality of patterns **82** for processing a packet received via the network **40**, and the storage unit **52** that stores information of the patterns **82** and the attribute information **62** related to the power reduction effect in association with each other. In the information processing system **2**, when a new pattern cannot be added to the storage unit **72** of the communication unit **30** due to a capacity shortage the storage unit **72**, the CPU **11** causes execution of the update processing of deleting the pattern **82** with the lowest power reduction effect based on the attribute information **62** and then adding the new pattern to the storage unit **72**.

[0096] Therefore, in the information processing system **2**, when a new pattern is added to the storage unit **72**, the pattern **82** with a high power reduction effect can be retained. Thus, the power consumption can be reduced by retaining the pattern **82** with a high power reduction effect in the information processing system **2**.

[0097] In the information processing system **2**, the CPU **11** acquires the rewrite count of the pattern **82**, and when the rewrite count is equal to or more than the specified value, the CPU **11** does not cause the execution of the update processing of adding the new pattern to the storage unit **72**. Therefore, in the information processing system **2**, it is possible to suppress a decrease in the service life of the storage unit **72**, compared with a case where the rewrite count of the pattern **82** is unlimited.

[0098] Furthermore, in the information processing system **2**, the attribute information **62** includes the usage record count of the pattern **82** and the packet processing time. The CPU **11** selects, based on the usage record count and the processing time, the pattern **82** with the lowest power reduction effect to be deleted. For this reason, in the information processing system **2**, the pattern **82** with a high power reduction effect is less likely to be deleted, compared with a case where the usage record count of the pattern and the packet processing time are not taken into consideration.

[0099] Additionally, in the information processing system **2**, the attribute information **62** includes the rewritability information on the pattern **82**, and the CPU **11** does not delete the pattern **82** of the storage unit **72** regardless of the power reduction effect when the pattern is unrewritable. Therefore, in the information processing system **2**, the pattern **82** that is unrewritable can be prevented from being deleted.

[0100] In the information processing system **2**, when the packet is a target of the processing time measurement, the CPU **11** measures the processing time of the packet and stores the processing time in the storage unit **52** of the main processing unit **10** as the attribute information **62**. Therefore, in the information processing system **2**, the pattern **82** with a higher power reduction effect is likely to be retained as compared with a case where the packet processing time is not taken into consideration.

[0101] In the information processing system **2**, the packet as the processing time measurement target is a packet involving no job. For this reason, in the information processing system **2**, the processing time is less likely to be measured for a packet the processing time for which varies depending on the job content.

[0102] The information processing system **2** further includes the communication unit **30** that performs packet communication via the network **40** and the CPU **31** provided in the communication unit **30**. The pattern **82** includes the discard pattern **82A** and the proxy response pattern **82B**. When a packet received by the communication unit **30** corresponds to the discard pattern **82A** or the proxy response pattern **82B**, the CPU **31** processes the packet without activating the CPU **11** of the main processing unit **10**. Therefore, in the information processing system **2**,

power consumption can be reduced, compared with a case where all the packets are processed by the CPU of the main processing unit.

[0103] Further, in the information processing system 2, after processing the packet, the CPU 31 stores the incremental information for incrementing the usage record count of the pattern 82 in the attribute information 62 in the storage unit 72, without activating the CPU 11 of the main processing unit 10. Therefore, the information processing system 2 can accumulate the attribute information 62 related to the power reduction effect.

[0104] In the information processing system 2, when the packet received by the communication unit 30 does not correspond to the discard pattern 82A or the proxy response pattern 82B, the CPU 31 outputs the usage record count of the pattern 82 to the CPU 11 in response to the request from the CPU 11 of the main processing unit 10. Thus, the CPU 11 causes the storage unit 52 to store the usage record count of the pattern 82 as the attribute information 62, whereby the attribute information 62 is accumulated. Therefore, in the information processing system 2, when the new pattern 82 is added to the storage unit 72, the pattern 82 with a high power reduction effect can be retained.

Others

[0105] The processing system of the present disclosure is not limited to the information processing system 2 described in the first exemplary embodiment, and various changes can be made. For example, each of the components, such as the image forming unit 22, the scan unit 24, and the fax machine 26 provided in the information processing system 2 can be changed. Further, the storage unit 72 is provided in the communication unit 30, but the present disclosure is not limited thereto, and the storage unit 72 may be configured by a RAM or the like connected to the communication unit 30.

[0106] The processing in the information processing system 2 described above can also be implemented by a dedicated hardware circuit. In this case, the processing may be executed by a plurality of units of hardware.

[0107] Further, the program for operating the information processing system 2 may be provided by a computer-readable recording medium such as a universal serial bus (USB) memory, a flexible disk, or a compact disc read only memory (CD-ROM), or may be provided online via a network such as the Internet. In this case, the program recorded in the computer-readable recording medium is generally transferred to and stored in a memory, a storage, or the like. In addition, the program may be provided as, for example, independent application software, or may be incorporated into software of each device as one function of the information processing system 2.

[0108] While a specific exemplary embodiment of the present disclosure has been described in detail, the present disclosure is not limited to such an exemplary embodiment, and it is apparent to those skilled in the art that various other exemplary embodiments are possible within the scope of the present disclosure.

APPENDIX

[0109] (((1)))

[0110] A processing system comprising: [0111] a first storage unit configured to store part of a plurality of processing procedures for processing a packet received via a network; [0112] a second storage unit configured to store information on the processing procedures and attribute information related to a power reduction effect in association with each other; and [0113] a first processor configured to execute, when a new processing procedure is not able to be added to the first storage unit due to a capacity shortage of the first storage unit, update processing of deleting one of the processing procedures the power reduction effect provided by which is lowest based on the attribute information and then adding the new processing procedure to the first storage unit. [0114] (((2)))

[0115] The processing system according to (((1))), wherein the first processor is configured to acquire a rewrite count of the processing procedure, and when the rewrite count is equal to or more

than a specified value, the first processor is configured not to execute the update processing of adding the new processing procedure to the first storage unit. [0116] (((3)))

[0117] The processing system according to (((1))) or (((2))), wherein [0118] the attribute information includes a usage record count of the processing procedure and a processing time for processing the packet, and [0119] the first processor is configured to select the processing procedure the power reduction effect provided by which is lowest to be deleted based on the usage record count and the processing time. [0120] (((4)))

[0121] The processing system according to any one of (((1))) to (((3))), wherein [0122] the attribute information includes rewritability information on the processing procedure, and [0123] the first processor is configured not to delete the processing procedure in the first storage unit regardless of the power reduction effect, when the processing procedure is unrewritable. [0124] (((5)))

[0125] The processing system according to any one of (((1))) to (((4))), wherein, when the packet is a target of measurement of the processing time, the first processor is configured to measure the processing time for the packet and store the processing time as the attribute information in the second storage unit. [0126] (((6)))

[0127] The processing system according to (((5))), wherein the packet that is the target of the measurement of the processing time is a packet involving no job. [0128] (((7)))

[0129] The processing system according to any one of (((1))) to (((6))), further comprising: [0130] a communication unit configured to perform communication for the packet via the network; and [0131] a second processor provided in the communication unit, wherein [0132] the processing procedures include a processing procedure of performing discarding or proxy response in the communication unit, and [0133] the second processor is configured to process the packet without activating the first processor when the packet received by the communication unit corresponds to the processing procedure of performing discarding or the proxy response. [0134] (((8)))

[0135] The processing system according to (((7))), wherein, after processing the packet, the second processor is configured to store, in the first storage unit, incremental information for incrementing a usage record count of the processing procedure in the attribute information, without activating the first processor. [0136] (((9)))

[0137] The processing system according to (((7))), wherein the second processor is configured to output a usage record count of the processing procedure to the first processor in response to a request from the first processor when the packet does not correspond to the processing procedure for performing the discarding or the proxy response. [0138] (((10)))

[0139] A processing method comprising executing by a computer, when a new processing procedure is not able to be added to a first storage unit configured to store part of a plurality of processing procedures for processing a packet received via a network due to a capacity shortage of the first storage unit, update processing of deleting one of the processing procedures power reduction effect provided by which is lowest, based on attribute information stored in a second storage unit configured to store information on the processing procedure and the attribute information related to the power reduction effect in association with each other, and then adding the new processing procedure to the first storage unit.

Claims

1. A processing system comprising: a first storage unit configured to store part of a plurality of processing procedures for processing a packet received via a network; a second storage unit configured to store information on the processing procedures and attribute information related to a power reduction effect in association with each other; and a first processor configured to execute, when a new processing procedure is not able to be added to the first storage unit due to a capacity shortage of the first storage unit, update processing of deleting one of the processing procedures the power reduction effect provided by which is lowest based on the attribute information and then

adding the new processing procedure to the first storage unit.

2. The processing system according to claim 1, wherein the first processor is configured to acquire a rewrite count of the processing procedure, and when the rewrite count is equal to or more than a specified value, the first processor is configured not to execute the update processing of adding the new processing procedure to the first storage unit.

3. The processing system according to claim 1, wherein the attribute information includes a usage record count of the processing procedure and a processing time for processing the packet, and the first processor is configured to select the processing procedure the power reduction effect provided by which is lowest to be deleted based on the usage record count and the processing time.

4. The processing system according to claim 1, wherein the attribute information includes rewritability information on the processing procedure, and the first processor is configured not to delete the processing procedure in the first storage unit regardless of the power reduction effect, when the processing procedure is unrewritable.

5. The processing system according to claim 1, wherein, when the packet is a target of measurement of the processing time, the first processor is configured to measure the processing time for the packet and store the processing time as the attribute information in the second storage unit.

6. The processing system according to claim 5, wherein the packet that is the target of the measurement of the processing time is a packet involving no job.

7. The processing system according to claim 1, further comprising: a communication unit configured to perform communication for the packet via the network; and a second processor provided in the communication unit, wherein the processing procedures include a processing procedure of performing discarding or proxy response in the communication unit, and the second processor is configured to process the packet without activating the first processor when the packet received by the communication unit corresponds to the processing procedure of performing discarding or the proxy response.

8. The processing system according to claim 7, wherein, after processing the packet, the second processor is configured to store, in the first storage unit, incremental information for incrementing a usage record count of the processing procedure in the attribute information, without activating the first processor.

9. The processing system according to claim 7, wherein the second processor is configured to output a usage record count of the processing procedure to the first processor in response to a request from the first processor when the packet does not correspond to the processing procedure for performing the discarding or the proxy response.

10. A processing method comprising: executing by a computer, when a new processing procedure is not able to be added to a first storage unit configured to store part of a plurality of processing procedures for processing a packet received via a network due to a capacity shortage of the first storage unit, update processing of deleting one of the processing procedures power reduction effect provided by which is lowest, based on attribute information stored in a second storage unit configured to store information on the processing procedure and the attribute information related to the power reduction effect in association with each other, and then adding the new processing procedure to the first storage unit.
